#DRY RUN

Example: $s = $ "rbb"
$t = $ "rb"

Expected ans = 2.

using memoization technique#

Logic: for match:
$dp[i][j] = dp[i-1][j-1] + dp[i-1][j]$

for no match.
$dp[i][j] = dp[i-1][j]$.

Step 1:
$s[1] = $'r' $==$ $t[1] = $'r']
$\Rightarrow dp[1][1] = dp[0][0] + dp[0][1] = 1+0 = 1$

Step 2:
$s[1] = $'r' $\neq$ $t[2] = $'b'
$\Rightarrow dp[1][2] = dp[0][2] = 0$

Step 3:
$s[2] = $'b' $\neq$ $t[1] = $'r' $\rightarrow dp[2][1] = d[1][1] = 1$

Step 4:
$s[2] = $'b' $==$ $t[2] = $'b' $\rightarrow dp[2][2] = dp[1][2] + dp[1][2] = 1+0 = 1$

Step 5: $s[3] = $'b' $\neq$ $t[1] = $'r' $\rightarrow dp[3][1] = dp[2][1] = 1$.

Step 6

S[3]='b' == t[2]='b' → dp[3][2] = dp[2][1] + dp[2][2] = 1+1 = 2.

dp Table.

	∅	r	b
∅	1	0	0
r	1	1	0
b	1	1	1
b	1	1	2

State space tree